Dynamical systems time series classification

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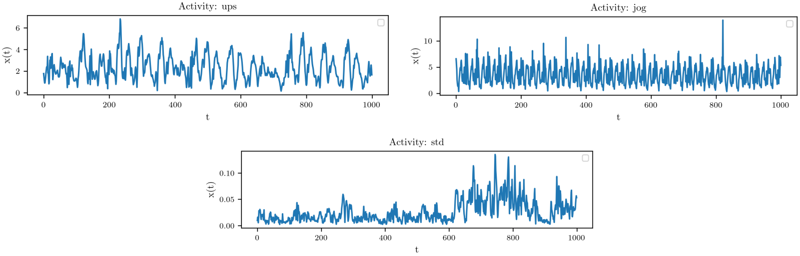
Problem

Time series classification originated from the K dynamical systems of the form

$$\begin{cases} \frac{d}{dt}\bar{\boldsymbol{x}}(t) = f(\bar{\boldsymbol{x}}), \\ \bar{\boldsymbol{x}}(0) = \bar{\boldsymbol{x}}_0, \end{cases} \tag{1}$$

assuming we have labeled sample series
$\mathcal{D} = \{(\boldsymbol{x}_1(t_i), y_1), \dots, (\boldsymbol{x}_N(t_i), y_N)\}$ where $\boldsymbol{x}(t_i) = \bar{\boldsymbol{x}}(t_i) + \epsilon(t_i)$.



Time series from different systems

Classification overview

1. Reconstruct (if needed) diffeomorphic phase trajectories from the observed time series: Whitney or Takens embedding theorem.

$$\{\tilde{\boldsymbol{x}}(t_i)\} \to \{\bar{\boldsymbol{x}}(t_i)\}$$

2. Learn (if not provided) vector fields of the systems $f_i(\boldsymbol{z})$ from the trajectories (NODE).

$$\hat{f} = \underset{f \in \mathcal{F}}{\arg\min}\ \mathcal{L}(\{\boldsymbol{x}(t_i)\}, f)$$

   Choose $\mathcal{L}, \mathcal{F}$ according to the prior knowledge of the system.

3. Associate a test trajectory with the system that gives least deviation from the *initial value problem* (IVP) (1).

$$k = \underset{k \in 1 \ldots K}{\arg\min}\ \rho\big(\{\boldsymbol{x}(t_i)\}, \mathrm{IVP}(t_i, f_k, \bar{\boldsymbol{x}}_0)\big)$$

Approach limitations

In steps 2-3 we need to compute precise IVP trajectories and know exact $\bar{\boldsymbol{x}}_0$. On practice, it is not possible for several reasons

1. In many cases IVP has no analytic solution and *numerical solvers* are used. They induce numerical error and may have stability issues.
2. Learned field $\hat{f}$ from step 2 is an approximation of the real f. It contributes to the IVP error.
3. Observed series $\boldsymbol{x}_{t_i}$ are noisy including starting point $\bar{\boldsymbol{x}}_0$.
4. The dynamical system (1) may not be an adequate time series model globally. For example, the SDE model may be more appropriate

$$\begin{cases} d\bar{\boldsymbol{x}}(t) = f(\bar{\boldsymbol{x}})dt + GdW(t), \\ \bar{\boldsymbol{x}}(0) = \bar{\boldsymbol{x}}_0, \end{cases} \tag{2}$$

Classification as the filtering problem

We use ODE solvers to approximate dynamics (1) and obtain the following state space model

$$\bar{\boldsymbol{x}}_{t_{i+1}} = \text{ODESolve}(\bar{\boldsymbol{x}}_{t_i})$$
$$\boldsymbol{x}_{t_i} = \bar{\boldsymbol{x}}_{t_i} + \epsilon_{t_i}$$
$$\bar{\boldsymbol{x}}_0 \sim F(x)$$

In this formulation we change $\text{IVP}(t_i, f_k, \bar{\boldsymbol{x}}_0)$ for $\mathbb{E}[\{\bar{\boldsymbol{x}}_{t_i}\}|\{\boldsymbol{x}_{t_i}, \bar{\boldsymbol{x}}_0\}]$ — the filtering problem. The classifier is now

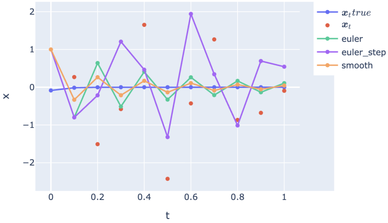
$$k = \operatorname*{arg\,min}_{k \in 1 \dots K} \rho\Big(\{\boldsymbol{x}(t_i)\}, \mathbb{E}[\{\bar{\boldsymbol{x}}_{t_i}\}|\{\boldsymbol{x}_{t_i}, \bar{\boldsymbol{x}}_0\}]\Big).$$

For the SDE dynamics we just obtain more general filtering problem

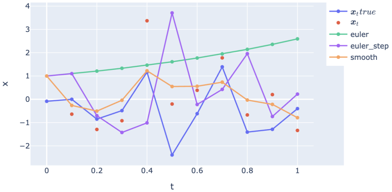
$$\bar{\boldsymbol{x}}_{t_{i+1}} = \text{ODESolve}(\bar{\boldsymbol{x}}_{t_i}) + \xi_{t_{i+1}}$$
$$\boldsymbol{x}_{t_i} = \bar{\boldsymbol{x}}_{t_i} + \epsilon_{t_i}$$
$$\bar{\boldsymbol{x}}_0 \sim F(x)$$

Experiments to justify filtering

Compare direct ODE solvers and filtering for the noisy trajectory of the ODE system $\dot{x} = x$



Compare direct ODE solvers and filtering for the noisy trajectory of the SDE system $dx_t = x_t(1 + dW_t)$



Выносится на защиту

1. Предложен метод классификации временных рядов в модели ODE/SDE с учётом случайного шума.
2. Получены теоретические результаты асимптотической оптимальности классификатора при достаточной информации о дин.системах и начальной точки. Обоснован подход с фильтрацией при отсутствии данных условий.
3. Подход эмпирически валидирован на синтетических и реальных данных.